

Systems Analysis and Design — Lab Sessions Agenda

Faculty of Computers and Information | Comprehensive 11-Lab Practical Schedule

Course Objective: Guide students step-by-step through practical engineering modules to build a professional System Documentation package.

Final Deadline: December 30, 2026

Core Output & Learning Outcomes

The ultimate deliverable of these practical lab sessions is a comprehensive System Documentation report. Students will apply systems analysis and design principles to model a complete software solution from scratch.

Documentation Deliverable: By the final deadline, each student group must submit a finalized System Documentation comprising administrative requests, economic evaluations, data flow models, unified modeling language (UML) specifications, and user interface mockups.

Integrated Practical Tools

The following tools are integrated into the 11 labs rather than taught as isolated topics: **Git/GitHub** for version control and collaborative file management; **Trello** as the primary beginner-friendly task management tool; **Jira** as an optional advanced Agile/Scrum project-management tool; **Draw.io** for DFD and UML diagrams; and **PlantUML** as an optional text-based UML alternative.

Lab	Topic	Tools	Practical Application	Assignment
01	Course Introduction & Project Team	GitHub + Trello/Jira	Create team, repository, project board and project folder structure.	Create GitHub Repository and Project Management Board.
02	System Request	Trello/Jira	Convert business needs and requirements into project tasks.	System Request + Project Board.
03	Methodology & Feasibility	Trello/Jira	Select Agile or Waterfall and organize project work accordingly.	Choose methodology and justify it.
04	Economic Feasibility Analysis	Excel + Trello/Jira	Record activities/costs and calculate ROI, BEP and NPV.	Economic feasibility analysis.
05	Staff Plan & Structure	Jira/Trello	Define roles, responsibilities and task ownership.	Team structure and task assignment.
06	Work Plan Development	Jira/Trello	Create WBS, milestones, tasks and timeline controls.	WBS and project timeline.
07	Functional & Non-functional Requirements	GitHub + Diagrams	Document requirements and prepare use-case-related models.	Requirements Specification.
08	Process Modeling: DFD (As-Is & To-Be)	Draw.io + GitHub	Create Context Diagram and Level 0 DFD for current and proposed workflows.	DFD As-Is and To-Be.
09	UML Diagrams Integration	Draw.io / PlantUML + GitHub	Create Use Case, Activity, Sequence and Class diagrams and map them to DFD processes.	Complete UML package.
10	User Interface Design	Figma / Draw.io	Design three manual UI mockups and screen layouts.	Three UI mockups.
11	Final Documentation & Submission	GitHub + Trello/Jira	Review, integrate and submit the complete project documentation.	Final System Documentation.

Tool Roles Across the Project

Git: Version control for project files and, where applicable, source code. Students record meaningful changes using commits and can recover previous versions.

GitHub: Shared online repository for the team project. It stores documentation, diagrams and other project artifacts and supports collaborative work.

Trello: Recommended primary task-management tool for beginners. Students organize work using To Do → In Progress → Review → Done.

Jira: Optional advanced project-management platform for Agile/Scrum practice, task assignment, milestones and workflow tracking.

Diagrams: Visual models used to analyze and design the system. Draw.io is recommended for DFD and UML; PlantUML may be introduced as an advanced text-based UML option.

Project Workflow

Plan → Analyze → Model → Manage → Version → Document

Consistency Requirement: DFD processes and UML models should remain aligned and cross-mapped. The final documentation should integrate requirements, feasibility, work planning, DFD, UML, UI and project records.

Evaluation & Submission Guidelines

Documentation Quality: All sections should be thoroughly written, professionally formatted, and logically structured.

Model Consistency: Ensure strict alignment and cross-mapping between DFD processes and UML structural/behavioral diagrams.

Deadline Adherence: Final System Documentation must be submitted on or before December 30, 2026.